Assignment: AI System Specification

Selected System: Smart Traffic Signal Controller

Task 3: Critical Analysis

1. Greatest Technical Challenge:
The dynamic environment is the greatest technical challenge for a Smart Traffic Signal Controller. Traffic conditions change continuously due to varying vehicle flow, pedestrians, accidents, and weather. The AI system must process real-time data and adapt instantly, which makes system design complex and challenging.

2. Utility Function:
U = 0.7 × traffic_flow_efficiency – 0.3 × average_waiting_time

If the weight of traffic_flow_efficiency is doubled, the system will prioritize clearing traffic more aggressively. This improves overall traffic flow but may increase waiting time for less busy lanes or pedestrians.

Structured Representation (JSON)


```
{ "system_name": "Smart Traffic Signal Controller", "peas": {
"performance_measures": [ "average_vehicle_waiting_time", "traffic_flow_efficiency",
"intersection_throughput", "accident_reduction_rate" ], "environment": "Urban road
intersections with vehicles, pedestrians, weather variations, and emergency
traffic", "actuators": [ "traffic_light_control_unit",
"pedestrian_signal_controller", "emergency_vehicle_priority_switch" ], "sensors": [
"vehicle_cameras", "inductive_loop_detectors", "pedestrian_buttons",
"weather_sensors" ] }, "environment_classification": { "observability": { "choice":
"Partially Observable", "justification": "Not all vehicles or pedestrian intentions
can be fully detected at all times." }, "agents": { "choice": "Multi-agent",
"justification": "Multiple traffic signals coordinate with nearby intersections." },
"determinism": { "choice": "Stochastic", "justification": "Traffic flow and driver
behavior are unpredictable." }, "episodic": { "choice": "Sequential",
"justification": "Each signal decision affects future traffic conditions." },
"dynamics": { "choice": "Dynamic", "justification": "Traffic conditions change
continuously in real time." }, "discreteness": { "choice": "Continuous",
"justification": "Traffic density and timing vary over a continuous range." } },
"utility_function": "U = 0.7 * traffic_flow_efficiency - 0.3 * average_waiting_time"
}
```